EEPROM CODES

1 BYTE

```
#include <EEPROM.h>

#define pulsePin 13

const int addr = 100;

void setup() {
  Serial.begin(115200);
  pinMode(pulsePin, OUTPUT);
}

void loop() {
  digitalWrite(pulsePin, LOW);
  delay(1000);

  Serial.println(F("1 BYTE EEPROM Test"));

  // WRITE TEST
  byte oldVal = EEPROM.read(addr);
  byte newVal = oldVal ^ 0xFF; // ^ = XOR to ensure real write

  digitalWrite(pulsePin, HIGH);
  unsigned long t0 = micros();
  EEPROM.write(addr, newVal);
  unsigned long t1 = micros();
```

EEPROM CODES

```
  digitalWrite(pulsePin, LOW);


  Serial.print(F("WRITE 1B time: "));
  Serial.print(t1 - t0);
  Serial.println(F(" us"));


  delay(1000); // Small gap before read


  // Read test
  digitalWrite(pulsePin, HIGH);
  t0 = micros();
  byte val = EEPROM.read(addr);
  t1 = micros();
  digitalWrite(pulsePin, LOW);


  Serial.print(F("READ 1B time: "));
  Serial.print(t1 - t0);
  Serial.println(F(" us"));
  Serial.print(F("Value read: "));
  Serial.println(val);


  Serial.println(F("Test complete."));
}
```

TWO BYTE

EEPROM CODES

```
#include <EEPROM.h>


#define pulsePin 13
#define SET_HI() digitalWrite(pulsePin, HIGH)
#define SET_LO() digitalWrite(pulsePin, LOW)


const int baseAddr = 200;


void setup() {
  Serial.begin(115200);
  pinMode(pulsePin, OUTPUT);
}


void loop() {
  digitalWrite(pulsePin, LOW);
  delay(1000);


  Serial.println(F("2 BYTE EEPROM TEST"));


  // WRITE TEST
  byte old1 = EEPROM.read(baseAddr);
  byte old2 = EEPROM.read(baseAddr + 1);
  byte new1 = old1 ^ 0xAA;
  byte new2 = old2 ^ 0x55;


  SET_HI();
```

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```
unsigned long t0 = micros();
EEPROM.write(baseAddr, new1);
EEPROM.write(baseAddr + 1, new2);
unsigned long t1 = micros();
SET_LO();

Serial.print(F("WRITE 2B time: "));
Serial.print(t1 - t0);
Serial.println(F(" us"));

delay(1000);

// READ TEST
SET_HI();
t0 = micros();
byte v1 = EEPROM.read(baseAddr);
byte v2 = EEPROM.read(baseAddr + 1);
t1 = micros();
SET_LO();

Serial.print(F("READ 2B time: "));
Serial.print(t1 - t0);
Serial.println(F(" us"));
Serial.print(F("Values: "));
Serial.print(v1);
Serial.print(F(", "));
```

EEPROM CODES

```
  Serial.println(v2);


  Serial.println(F("Test complete."));
}
```

FOUR BYTE

```
#include <EEPROM.h>


#define pulsePin 13
#define SET_HI() digitalWrite(pulsePin, HIGH)
#define SET_LO() digitalWrite(pulsePin, LOW)


const int baseAddr = 300;


void setup() {
  Serial.begin(115200);
  pinMode(pulsePin, OUTPUT);
}


void loop() {
  digitalWrite(pulsePin, LOW);
  delay(1000);


  Serial.println(F("4 BYTE EEPROM TEST"));
```

EEPROM CODES

```
// WRITE TEST
byte vals[4];

SET_HI();
unsigned long t0 = micros();
for (int i = 0; i < 4; i++){
  EEPROM.write(baseAddr + i, vals[i] ^ (0x0F + i));
}
unsigned long t1 = micros();
SET_LO();

Serial.print(F("WRITE 4B time: "));
Serial.print(t1 - t0);
Serial.println(F(" us"));

delay(1000);

// READ TEST
SET_HI();
t0 = micros();
for (int i = 0; i < 4; i++){
  vals[i] = EEPROM.read(baseAddr + i);
}
t1 = micros();
SET_LO();
```

EEPROM CODES

```
  Serial.print(F("READ 4B time: "));
  Serial.print(t1 - t0);
  Serial.println(F(" us"));

  Serial.println(F("Test complete."));
}
```

FULL BYTE

```
#include <EEPROM.h>

#define pulsePin 13
#define EEPROM_SIZE 4096

void setup() {
  Serial.begin(115200);
  pinMode(pulsePin, OUTPUT);
}

void loop() {
  digitalWrite(pulsePin, LOW);

  Serial.println(F("EEPROM FULL SWEEP TIMING"));

  // WRITE TEST
  digitalWrite(pulsePin, HIGH);
```

EEPROM CODES

```
  unsigned long t0 = micros();
  for (int i = 0; i < EEPROM_SIZE; i++) {
    EEPROM.write(i, i % 256);  // A changing pattern
  }
  unsigned long t1 = micros();
  digitalWrite(pulsePin, LOW);
  Serial.print(F("WRITE full (4096B): "));
  Serial.print(t1 - t0);
  Serial.println(F(" us"));

  // READ TEST
  digitalWrite(pulsePin, HIGH);
  t0 = micros();
  for (int i = 0; i < EEPROM_SIZE; i++) {
    volatile byte data = EEPROM.read(i);
  }
  t1 = micros();
  digitalWrite(pulsePin, LOW);
  Serial.print(F("READ full (4096B): "));
  Serial.print(t1 - t0);
  Serial.println(F(" us"));

  Serial.println(F("DONE"));

}
```